

SAT UZBEKISTAN

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1.

A company has a customer loyalty program. In January 2018, there were 900 customers enrolled in the loyalty program. For the next 24 months after January 2018, the total number of customers enrolled in the loyalty program each month was 2% greater than the total number enrolled the previous month. Which equation gives the total number of customers, c , enrolled in the company's loyalty program m month after January 2018, where $m \leq 24$?

A) $c = 900(0.02)^m$

B) $c = 900(1.02)^m$

C) $c = 900(1.2)^m$

D) $c = 900(2)^m$

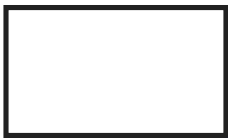
2.

The rectangular region of land is divided into 40 square lots of equal area R , in square units. The length of the region is 1.6 times the width of the region. If the width of the region is $x\sqrt{R}$ units, what is the value of x ?



3.

A rectangular area consists of 3,179 equal squares where each square has an area of k . If the length of the rectangular area is 2.75 times the width and the width is equal to $x\sqrt{k}$, what is the value of x ?



4.

A rectangular area consists of 1,372 equal squares where each square has an area of k . If the width of the rectangular area is 1.75 times the length and the length is equal to $x\sqrt{k}$, what is the value of x ?

5.

A circle in the xy -plane has its center at $(-3, 3)$. Line t is tangent to this circle at the point $(6, -3)$. Which of the following points also lies on line t ?

- A. $(0, \frac{3}{2})$
- B. $(3, 12)$
- C. $(12, 6)$
- D. $(15, 3)$

6.

Which of the following has $x + 2a$ as a factor where a is a positive integer?

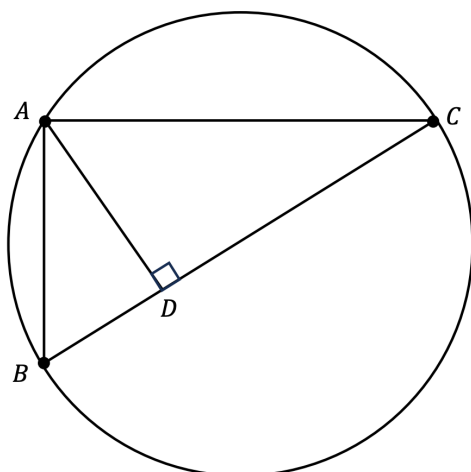
- A) $3x^2 + 7x + 14a = 0$
- B) $3x^2 + 28x + 14a = 0$
- C) $3x^2 + 42x + 14a = 0$
- D) $3x^2 + 49x + 14a = 0$

7.

An equilateral triangle has a perimeter of 1200 meters. If the height of the triangle is equal to $x\sqrt{3}$, what is the value of x ?

8.

In the given figure, BC is the diameter of the circle. If the length of BC is equal to 132 and the length of AB is equal to $\sqrt{363}$, what is the value of $\frac{BC}{BD}$?



9.

x	$f(x)$
-9	155
-3	227
3	155

Three points on the quadratic function $f(x)$ are given in the table. If $g(x) = f(x + 4)$, what is the y -intercept of the function $g(x)$?

10. A savings account is opened with an initial deposit of \$6,000. The amount of money in the account t years after the initial deposit is given by the function $f(t) = 6,000(1.01)^{2t}$. Which of the following is the best interpretation of the statement " $f(5)$ is approximately equal to 6,627.73" in this context?

- A) 5 years after the initial deposit, the amount of money, in dollars, in the account is 6,627.73
- B) Every 5 years, the amount of money, in dollars, in the account increases by 6,627.73
- C) 5 years after the initial deposit, the amount of money, in dollars, in the account has increased by 6,627.73
- D) Every 5 years, the amount of money, in dollars, in the account decreases by 6,627.73

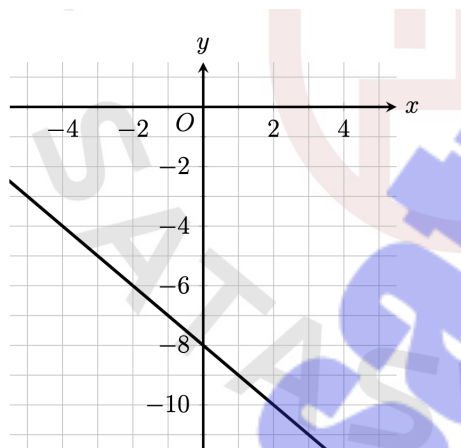
11.

$$x^2 - 12x + c = 0$$

In the given equation, c is a constant. The equation has no real solutions if $c > n$. What is the least possible value of n ?

- A) -36
- B) -12
- C) 12
- D) 36

12.



The graph of the linear function $y = f(x) - 19$ is shown. If c and d are positive constants, which equation could define f ?

- A) $f(x) = d - cx$
- B) $f(x) = -d + cx$
- C) $f(x) = d + cx$
- D) $f(x) = -d - cx$

13.

A circle has center at point $(5, 7)$ and a tangent line (to this circle) with slope $\frac{5}{6}$ that passes through $(a, -4)$. Find the value of a .

- A. $-\frac{25}{6}$
- B. $\frac{85}{6}$
- C. $\frac{91}{6}$
- D. $-\frac{6}{91}$

14.

The surface area of the square pyramid is $100 + 20\sqrt{146}$

and it includes base area. If base area equals to 100, what is the height of the pyramid?

15.

For 100 neurons, the table summarizes the distribution of classification and cell body diameter:

Classification	Cell body diameter (micrometers)		
	Less than 20	20 to 30	Greater than 30
Sensory neuron	13	7	3
Motor neuron	0	15	18
Interneuron	10	34	0

One of these neurons will be selected at random. What is the probability of selecting a neuron with a cell body diameter that is less than or equal to 30 micrometers, given that it is not classified as a motor neuron?

16.

The equation $h = \frac{9(v - 273.15)}{5} + 32$ gives the corresponding temperature h , in degrees Fahrenheit, of any substance that has temperature of v kelvins, where $v \geq 0$. If a substance has a temperature of 467.33 degrees Fahrenheit, what is the corresponding temperature, in kelvins, of this substance?

17.

$$F(x) = \frac{9}{5}(x - 273.15) + 32$$

The function F gives the temperature, in degrees Fahrenheit, that corresponds to x kelvins. If a temperature is increased by 11.70 kelvins, by how much did the temperature increase in degrees Fahrenheit?

18.

In the linear function f , $f(0) = 2$ and $f(7) = 2$. Which equation defines f ?

☐ (A) $f(x) = 0$

☐ (B) $f(x) = 7$

☐ (C) $f(x) = 2$

☐ (D) $f(x) = x + 2$

19.

Which of the following lists represents a data set with the smallest standard deviation?

☐ (A) 44, 45, 47, 49, 50

☐ (B) 45, 45, 47, 49, 49

☐ (C) 45, 46, 47, 48, 49

☐ (D) 46, 47, 47, 47, 48

20.



Note: Figure not drawn to scale.

The triangle shown has one side with a length of x inches and two sides each with a length of y inches. The perimeter of the triangle is 47 inches. Which equation represents this situation?

☐ (A) $x + y = 47$

☒ (B) $x + 2y = 47$

☐ (C) $2x + y = 47$

☐ (D) $2x + 2y = 47$

21.

$$W(x) = 2x + 60$$

The function W gives the total time, in minutes, it takes for a gardener to water x plants and mow the grass. According to the function, how many minutes does it take the gardener to mow the grass?

22.

Data set A consists of 10 positive integers less than 60. The list shown gives 9 of the integers from data set A.

43, 45, 44, 43, 38, 39, 40, 46, 40

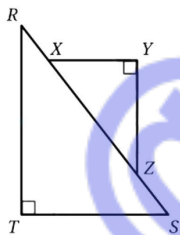
The mean of these 9 integers is 42. If the mean of data set A is an integer that is greater than 42, what is the value of the largest integer from data set A?

23.

x	y
20	-20
24	12
28	-20

The table shows three values of x and their corresponding values of y , where $y = f(x) + 6$ and f is a quadratic function. What is the y -coordinate of the y -intercept of the graph of $y = f(x)$ in the xy -plane?

24.



Note: Figure not drawn to scale.

In triangles RST and XYZ shown, $\overline{XY}$ is parallel to $\overline{TS}$ and $\tan R = \frac{180}{299}$. What is the value of $\sin X$ in triangle XYZ ?

25.

$$7x + 8y = 3$$

The given equation is one equation in a system of two linear equations. If the system of equations has at least one solution, which of the following equations could be the other equation in the system?

I. $10.5x + 12y = 4.5$

II. $10.5x - 12y = 4.5$

- A. II only
B. Neither I nor II
C. I and II
D. I only

26.

The function q is defined by

$$q(x) = a|x - 9|^2 - 53|x - 9| + b$$

where a and b are constants, and $a > b > 53$. If $q(539) = h$ and $q(-521) = k$, where h and k are constants, what is the value of

$$32(-539)^{h-k} + 347(9)^{k-h}?$$



27.

A circuit has 10 resistors connected in series with a total resistance of 200 ohms. The total resistance of 4 of these resistors is 80 ohms. Which inequality best represents all possible values of the resistance x , in ohms, of one of the other 6 resistors?

- A) $0 < x < 10$
B) $0 < x < 120$
C) $120 < x < 200$
D) $80 < x < 200$

28.

A shopper has \$30 to spend on pencils and notebooks. Each pencil costs \$1.50, and each notebook costs \$3.20. Let p be the number of pencils and n the number of notebooks. Which inequality shows how many items the shopper can buy?

- A) $1.5p + 3.2n \leq 30$
B) $1.5p + 3.2n \geq 30$
C) $3.2p + 1.5n \leq 30$
D) $3.2p + 1.5n \geq 30$

29.

A dance team aims to raise \$1,500 by selling show tickets. They earn \$15 per ticket. Which inequality shows the number of tickets t they need to sell?

- A) $t + 15 \leq 1,500$
B) $t + 15 \geq 1,500$
C) $15t \leq 1,500$
D) $15t \geq 1,500$

30.

$$\begin{aligned}y &> 20 \\ 3x + y &< 30\end{aligned}$$

If the point $(x, 25)$ is a solution, what could be the value of x ?

- A) 1
- B) 3
- C) 5
- D) 8

31.

$$\begin{aligned}y &< 35 - 5x \\ \frac{y}{5} &> 7\end{aligned}$$

Which inequality represents the x -values for all solutions (x, y) that satisfy the system?

- A) $x > 4$
- B) $x < 4$
- C) $x > -2$
- D) $x < 0$

32.

A freelance graphic designer charges **\$25 per hour** for the first **30 hours** worked in a month. For every hour worked **beyond 30 hours**, she charges **\$40 per hour**. What is the **minimum number of hours** she must work in a month to earn **at least \$1,150**?

- A) 39
- B) 40
- C) 41
- D) 42

33.

An employee earns **\$20 per hour** for the first **35 hours** worked in a week. For each **additional hour** worked beyond 35, the employee earns **\$30 per hour**. What is the **least number of whole hours** the employee must work in a week to earn **at least \$980**?

- A) 43
- B) 44
- C) 45
- D) 46

34.

A baby elephant weighed 800 pounds at birth and gained 20 pounds per day. If its weight after n days was more than 1,000 pounds but less than 1,200 pounds, what is one possible value of n ?

35.

In a set of four consecutive even integers, the first is represented by x . The product of 12 and the third even integer is at most the value of 30 less than the sum of the first and fourth even integers. What is the greatest possible value of x ?

36.

In a set of four consecutive odd integers, the smallest is represented by x . The sum of 15 and **twice** the second odd integer is **no more than** the difference between the fourth and the first odd integers **plus 50**. What is the greatest possible value of x ?

37.

A small business owner has a budget of **\$2,200** to purchase candles. To qualify for discounted pricing, they must buy at least **200** candles. The cost per small candle is **\$4.90**, while each large candle costs **\$11.60**. What is the maximum number of large candles the owner can purchase while staying within budget and meeting the minimum purchase requirement?

38.

$$-7(5x - 2)^2 + 3(5x - 1)^2$$

The expression above can be rewritten in the form

$$\frac{r}{4}x^2 + \frac{s}{4}x + \frac{t}{4}, \text{ where } r, s, \text{ and } t \text{ are constants.}$$

What is $r + s + t$?

- A) -80
- B) -60
- C) -40
- D) -20

39.

The quadratic function

$$ax^2 + 40x + c$$

has **at least one real solution**, where a and c are positive integers. What is the greatest possible value of ac ?

- A) 361
- B) 380
- C) 400
- D) 420

40.

In the given equation, m and n are constants, where $m > 4n$:

$$(x - m)^2 = (m - 4n)(x - m).$$

The sum of the solutions to this equation is $3m + 15$. What is the value of n ?

41.

For the parabola $px^2 + qx + r$, the x -intercepts are $(10, 0)$ and $(m, 0)$.

If $f(25) = f(3)$, what is the value of m ?

- A) 16
- B) 18
- C) 20
- D) 22

42.

The function $g(z) = pz^2 + qz + r$ has zeros at $z = 1$ and $z = 7$.

Suppose p is an integer greater than 1. Which of the following could be the value of $p + q$?

- A) -14
- B) -16
- C) -8
- D) -24

43.

When the quadratic function h is graphed in the xy -plane, where $y =$

$h(x)$, its vertex is $(2, -3)$. One of the x -intercepts of this graph is

$(\frac{11}{4}, 0)$. What is the other x -intercept?

- A) $(\frac{3}{4}, 0)$
- B) $(-1, 0)$
- C) $(\frac{5}{4}, 0)$
- D) $(-\frac{5}{4}, 0)$

44.

A quadratic function models the height, in meters, of an object launched from an elevated surface. Measurements show that at $t = 8$ s, the object is 254 m high, and at $t = 12$ s, it is 270 m high.

If the object's height at launch ($t = 0$) was 30 m, what is its height at $t = 17$ s?

- A) 140
- B) 180
- C) 200
- D) 220